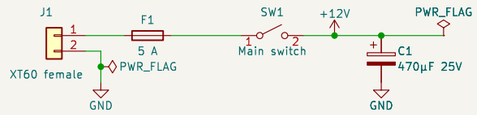
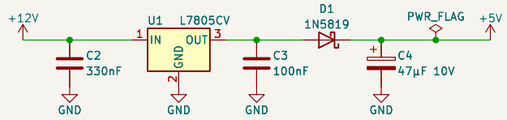
Power input, fuse, main switch



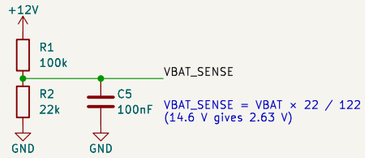
J1: female XT60 on the case, for the battery (LiFePO4 12.8 V 6 Ah with BMS) or a lab supply (12–14 V, current limit about 2 A).
The battery is charged off the mount.

5 V supply for the ESP32 board



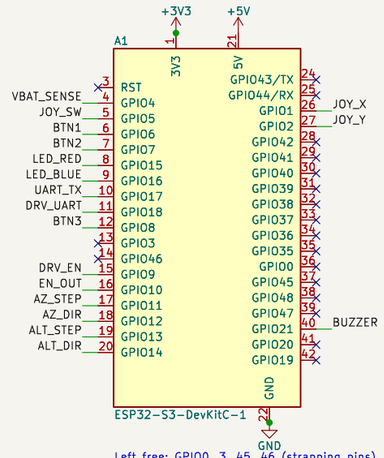
U1 on a small TO-220 heatsink (tab = GND), about 1 W to dissipate.
D1 stops the board's USB 5 V from feeding U1.

Battery voltage



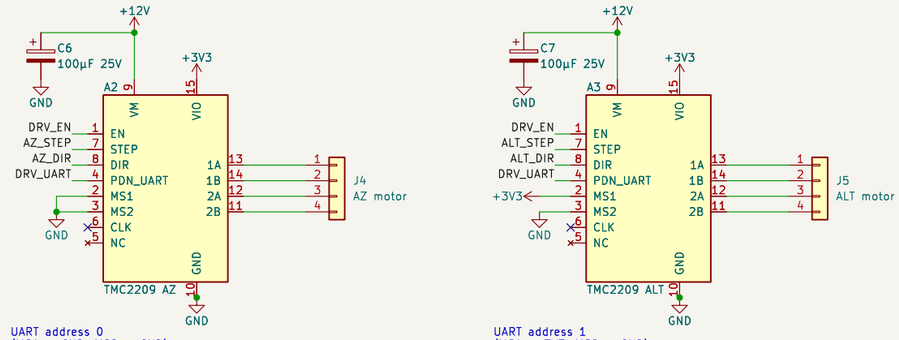
VBAT_SENSE = VBAT × 22 / 122
(14.6 V gives 2.63 V)

Controller



Left free: GPIO0, 3, 45, 46 (strapping pins), GPIO19/20 (USB), GPIO35–37 (PSRAM on some boards), GPIO43/44 (USB serial console).

Stepper drivers



UART address 0
(MS1 = GND, MS2 = GND)

UART address 1
(MS1 = 3V3, MS2 = GND)



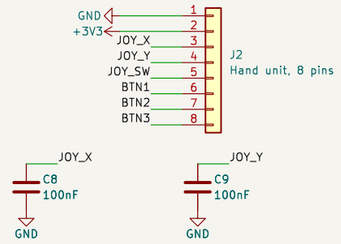
Both drivers share one single-wire UART:
GPIO17 (TX) through R4, GPIO18 (RX) direct.

E-stop (on the case)



GPIO10 drives EN_OUT low to enable the drivers.
E-stop pressed or ESP32 in reset: R3 pulls DRV_EN high and both drivers switch off.
GPIO9 reads DRV_EN to detect it.

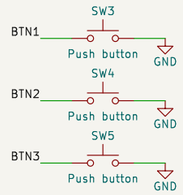
Hand-unit socket (on the case)



Hand unit (on the cable)

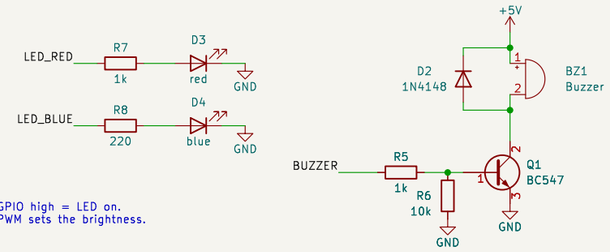


Analog joystick module (GND, VCC, VRx, VRy, SW), powered from 3.3 V.



Buttons and the joystick switch use the ESP32 internal pull-ups.

LEDs and buzzer (on the case)



GPIO high = LED on.
PWM sets the brightness.

Notes

1. Supply, controller and drivers are all on the L: no cable crosses the azimuth joint.
2. +12V is the switched supply rail (12.0–14.6 V). It feeds the drivers and U1.
3. TMC2209 modules: VIO at 3.3 V (ESP32 logic level); motor current is set over UART. The header pin order differs between makers: wire by pin name.
4. Motor connectors: pins 1–2 = one coil, pins 3–4 = the other coil.
5. A1 pin numbers: 1–22 = header J1, 23–44 = header J3 of the DevKitC-1.
6. The joystick and buttons are in a hand unit on an 8-wire cable (J2: for example a GX16-8 socket, or RJ45 with a network cable).
7. BZ1: active buzzer (GPIO21 high = sound) or passive buzzer (PWM tone on GPIO21). D2 protects Q1 from a magnetic buzzer's coil.
8. Blue light spoils dark adaptation more than red: keep the blue LED dim, or use it only for setup.

All electronics on the L of the mount
ESP32-S3 controller, two TMC2209 drivers, 12.8 V LiFePO4 battery or lab supply

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Title: Star-Track mount electronics

Size: A3	Date: 2026-10-25	**Rev: 2**
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